THE SPHERICAL REGGE SCISSORS THEOREM: A CONDITIONAL PROOF THROUGH AN EXPLICIT WEIGHT-TWO CONTRACT

Abstract. We give a conditional proof that every nondegenerate spherical tetrahedron is scissors congruent to each elementary Regge mate. The hypotheses are an explicit theorem package separating published field-level results from the relative motivic and comparison bridge still required over the rational parameter space. Internally we derive the monomial Regge action on angle phases, rationalize the orientation double cover, prove the finite four-channel tensor cancellation, and carry out the primitive-kernel and scissors-group diagram chases. We explain the relative quadric motive that the external package is intended to supply, including its three weight pieces, coproduct, Artin descent, and fibre comparison, without treating those assertions as consequences of ordinary homology. An accompanying Lean development checks the internal algebra and conditional implication without project axioms; it represents every motivic and scissors-theoretic input by a named field of the proof contract.

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1. Statement, proof contract, and logical route

1.1. **Spherical scissors congruence.** Let $S^3 \subset \mathbb{R}^4$ be the unit sphere. A spherical polytope is a finite union of geodesic simplices contained in open hemispheres. The spherical scissors group $\widehat{\mathcal{P}}(S^3)$ is generated by symbols $[P]$, one for each such polytope, subject to

$$[P] = [P_1] + \cdots + [P_m]$$

when the P_i have disjoint interiors and union P, and $[P] = [gP]$ for every spherical isometry g, including reflections. This is the unoriented geometric scissors group used in the conclusion; the orientation of a complex quadric introduced later is auxiliary data, carried through the parity line in Input (E6). Equality in this group is the stable algebraic form of equidecomposability. The cancellation theorem listed in Input (E8) below turns the equality obtained at the end into a finite equidecomposition.

Label the vertices of a nondegenerate spherical tetrahedron by $0, 1, 2, 3$, and select the opposite edge pair $01, 23$. The elementary Regge operation fixes those two edge lengths and replaces the remaining four lengths $\ell_1, \dots, \ell_4$ by

$$\ell_i^R = \sigma_\ell - \ell_i, \qquad \sigma_\ell = \frac{1}{2}\sum_{j=1}^{4} \ell_j. \tag{1.1}$$

Input (E1) says that the same transformation holds for the four moving dihedral angles and that the two tetrahedra have equal volume.

Theorem 1.1 (Conditional spherical Regge scissors theorem)**.** *Assume Inputs* (E1)–(E8) *below, including the relative motivic bridge in Inputs* (E2)–(E5)*, for the two universal rational charts of Section 2. If $T \subset S^3$ is a nondegenerate*

spherical tetrahedron and T^R is a nondegenerate elementary Regge mate, then

$$[T] = [T^R] \qquad \text{in } \widehat{\mathcal{P}}(S^3).$$

Consequently T and T^R admit a finite spherical equidecomposition.

No algebraicity hypothesis is imposed on the angles or vertices. The qualifier *conditional* is essential: this manuscript proves the implication from the following precisely typed inputs; it does not claim that the relative motivic bridge has already been proved in the cited literature.

1.2. **The external theorem package.** We separate published theorems from the additional comparison statements needed to use them in a family. This separation is also the interface used by the Lean formalization.

(E1) **Regge geometry.** For a genuine spherical Regge pair, the same linear involution acts on the four moving edge lengths and on the four moving dihedral angles, while the chosen opposite pair is fixed; the two tetrahedra have equal volume. This is imported from Akopyan–Izmestiev [AI19].

(E2) **Relative quadric coefficients.** For each rational chart U constructed in Section 2, the oriented quadric families attached to G and G^R determine elements

$$c_U(G), c_U(G^R) \in A_{02}(U)$$

of a rational vector space, and there is a linear reduced coproduct

$$\overline{\Delta}_U : A_{02}(U) \longrightarrow A_{11}(U).$$

These data are functorial under pullback to a complex point. Their intended construction is the framed relative face motive described in Section 3; existence and functoriality over this positive-dimensional base are input, not an internal consequence of the matrix calculation.

(E3) **Three-weight and primitive comparison.** There is a rational vector space H_U, an identification

$$\ker \overline{\Delta}_U \xrightarrow{\sim} H_U, \qquad H_U = H^1(U, \mathbb{Q}(2)),$$

and $H_U = 0$. The intended proof combines a three-step Artin–Tate filtration, a heart-to-derived Ext^1 comparison, weight-two birational invariance, and Borel's rank theorem. Garkusha [Gar23] and Borel [Bor77] supply the cited birational and number-field ingredients. The assertion that the particular relative face objects lie in a strict heart with this primitive kernel is part of the input.

(E4) **Relative coproduct formula and descent.** The coproduct of a framed quadric coefficient is the sum of its six oriented length–angle channels,

$$\overline{\Delta}_U c_U(G) = \sum_e L_e \otimes A_e,$$

and similarly for G^R. Five channels are ordinary Tate channels and the fixed p-channel may be a sign-Artin channel. Pullback to a splitting

cover, the normalized twisted trace, and the displayed formula are compatible, so equality after pullback descends to U. Pullback to every complex point commutes with the reduced coproduct and identifies the pulled back six channels with Goncharov's field-level coproduct; in particular, a universal zero-coproduct defect specializes to a primitive quadric-scissors class. This input also includes the algebraic recovery of the length phases and the Zariski-density step which extends the real Regge relations to that splitting cover. Goncharov's field-level coproduct calculation [Gon99] motivates this input; its exact relative and Artin-trace form is not proved here.

(E5) **Fibre comparison.** For every complex point $s : \operatorname{Spec}\mathbb{C} \to U$, specialization of relative coefficients and passage to the endpoint are linear maps. Let $\delta_s^\vee$ be the dual quadric-scissors defect. If $\delta_U^\vee = c_U(G) - c_U(G^R)$, then the endpoint obtained from $s^*\delta_U^\vee$ is exactly

$$c_G(\delta_s^\vee) \in \operatorname{gr}_2^\gamma K_3(\mathbb{C}) \otimes \mathbb{Q},$$

with the same signs, projective duality, frames, and Tate twist as in Goncharov's definition. This is an equality of classes, not merely of regulators. The family-to-fibre identification is an explicit input.

(E6) **Goncharov's field-level injections.** The canonical map from primitive weight-two quadric-scissors classes to $\operatorname{gr}_2^\gamma K_3(\mathbb{C}) \otimes \mathbb{Q}$ is injective, and the spherical map from the reduced spherical scissors group to the spherical part of the quadric group is injective. We use the projective-duality involution as well, with the explicit generator compatibility taking the spherical defect represented by $Q_{G^{-1}}$ to the dual defect represented by Q_G. These are the precise consequences of Goncharov's Theorems 4.13 and 4.14 used below [Gon99].

(E7) **Spherical presentation comparison.** The reduced geometric spherical scissors group is the spherical group on which Input (E6) is defined. After rationalization the comparison is an isomorphism and carries the particular reduced Regge defect to the particular Goncharov spherical defect used by Input (E6). This is the comparison with the coinvariant configuration presentation supplied by Dupont's Theorem 7.14 [Dup01].

(E8) **Elementary spherical scissors facts.** The groups $\widehat{\mathcal{P}}(S^2)$ and $\widehat{\mathcal{P}}(S^3)$ are uniquely divisible. The reduction map to $\widetilde{\mathcal{P}}(S^3) = \operatorname{coker}\Sigma$ is additive and exact at $\widehat{\mathcal{P}}(S^3)$: a class has zero reduction exactly when it is a suspension. The canonical rationalization map on $\widetilde{\mathcal{P}}(S^3)$ is injective. Area and volume are additive, area identifies $\widehat{\mathcal{P}}(S^2)$ with $\mathbb{R}$, and equality in $\widehat{\mathcal{P}}(S^3)$ implies finite equidecomposability by Zylev cancellation (with Gerling's orientation refinement). We also take as input the normalization

$$\operatorname{Vol}(\Sigma Q) = \frac{\pi}{2} \operatorname{Area}(Q)$$

for spherical suspension. The scissors statements are imported in the form recorded by Dupont [Dup01]; the normalization is the warped-product volume calculation displayed in Section 5.

Inputs (E2)–(E5) are deliberately stronger and more explicit than the slogan *motivic formalism is standard*. The cited sources develop the field-level constructions and the ingredients suggested above, but this manuscript does not supply a reference proving their exact simultaneous relative form. Until that bridge is constructed, the result of this paper is the conditional theorem stated above.

1.3. **What Lean checks.** The accompanying Lean development does not define Voevodsky motives, motivic cohomology, or algebraic K-theory. Instead it represents Inputs (E1)–(E8) by explicit structures and proves, without additional axioms, the implication from an inhabitant of those structures to the final scissors equality. It also checks the finite algebra that is specific to Regge symmetry: the involution and orthogonality of the Regge matrix, its monomial phase action, the coordinate identities and covering criterion for the two orientation charts, the fixed quadratic deck transformation, the tensor cancellation, the primitive-kernel deduction, and the final diagram chase. The exact manuscript-to-declaration map is in `formal/MANUSCRIPT_MAP.md`. The scheme-level principal-open identifications and finite-etaleness statements are standard external algebraic-geometry packaging and are marked as mixed rather than kernel-checked items in that map.

This distinction is important. A Lean theorem universally quantified over an external package is a verified conditional theorem; it is not a formal construction of that package and does not certify the cited analytic, algebro-geometric, or motivic results.

1.4. **Route of the conditional proof.** The proof has seven steps.

1. Replace each dihedral angle θ by its phase $u = e^{\mathrm{i}\theta}$. The additive Regge formula becomes the monomial substitution $u_i^R = \tau/u_i$.
2. Adjoin the orientation root $w = \sqrt{\det G}$. The substitutions $r = w - sx$ and $r_+ = w + sx$ rationalize the two necessary charts. Every physical pair gives a complex point of one of the resulting opens U.
3. Apply Input (E2) to form $\delta_U^\vee = c_U(G) - c_U(G^R)$. Section 3 explains the relative face motive which this symbol is intended to denote.
4. Use Input (E4) to translate its coproduct into six length–angle tensors. The two fixed tensors cancel directly; orthogonality of the Regge matrix cancels the other four. The descent clause handles the single possible sign-Artin channel.
5. Thus $\delta_U^\vee \in \ker \overline{\Delta}_U$. Input (E3) identifies this kernel with the zero group $H^1(U, \mathbb{Q}(2))$, so $\delta_U^\vee = 0$.
6. Pull back to the point of the original tetrahedron. Inputs (E5)–(E8) and the stated comparison and rationalization injections give zero in the reduced spherical scissors group.

7. The unreduced difference is a suspension. Regge volume equality, the suspension-volume formula, and Input (E8) kill that suspension and produce a finite equidecomposition.

Section 2 starts with the fully explicit algebra in Steps 1–2.

2. From spherical angles to rational orientation charts

This section replaces the trigonometric Regge transformation by an algebraic family. There are three steps. First, exponentiating the dihedral angles turns the linear Regge transformation into a monomial involution of an algebraic torus. Second, adjoining a square root of the Gram determinant retains the orientation of the associated quadric. Third, a change of variable rationalizes this double cover. The only angle phase which is lost in the rationalization is recovered by one explicit quadratic finite étale cover.

No motivic input is used in this section.

2.1. **The spherical angle Gram matrix.** Let $T \subset S^3$ be a nondegenerate spherical tetrahedron contained in an open hemisphere. Write its four facets as $F_0, \dots, F_3$, and let $n_i \in S^3 \subset \mathbf{R}^4$ be the outward unit normal to F_i. If F_i and F_j meet at interior dihedral angle $\theta_{ij} \in (0, \pi)$, then their outward normals meet at the supplementary angle. Hence

$$n_i \cdot n_j = -\cos\theta_{ij}. \tag{2.1}$$

The geometric edge at which these facets meet is indexed by the complementary pair of vertices. We use the facet-pair labels ij, since they are the natural labels for the Gram matrix. Complementing all six labels preserves opposite pairs and merely changes the edge-label convention.

Definition 2.1 (Angle Gram matrix)**.** The angle Gram matrix of T is

$$G(T) = (g_{ij})_{0 \le i,j \le 3}, \qquad g_{ii} = 1, \qquad g_{ij} = -\cos\theta_{ij} \quad (i \ne j). \tag{2.2}$$

If N is the matrix with columns $n_0, \dots, n_3$, then $G = N^t N$. In particular, a nondegenerate spherical tetrahedron has

$$G > 0, \qquad \det G = (\det N)^2 > 0. \tag{2.3}$$

The positivity is used only to locate the physical real locus. All algebraic constructions below allow the entries of G to be complex.

Fix the opposite Gram positions 01 and 23. In the cyclic order

$$(01, 02, 12, 23, 13, 03), \tag{2.4}$$

write the two fixed angles as θ_{01}, θ_{23} and the four moving angles as

$$\boldsymbol{\theta} = (\theta_{02}, \theta_{12}, \theta_{13}, \theta_{03})^t. \tag{2.5}$$

Put $\mathbf{1} = (1,1,1,1)^t$, let $J_4 = \mathbf{1}\mathbf{1}^t$, and define

$$R = \frac{1}{2} J_4 - I_4. \tag{2.6}$$

If

$$S = \frac{1}{2}(\theta_{02} + \theta_{12} + \theta_{13} + \theta_{03}), \tag{2.7}$$

then

$$R\boldsymbol{\theta} = (S - \theta_{02}, S - \theta_{12}, S - \theta_{13}, S - \theta_{03})^t. \tag{2.8}$$

Lemma 2.2 (Elementary properties of the Regge matrix)**.** *The matrix R is a symmetric orthogonal involution:*

$$R^t = R, \qquad R^2 = I_4.$$

It is the identity on $\mathbf{Q1}$ *and minus the identity on* $\mathbf{1}^\perp$*.*

Proof. Since $J_4^2 = 4J_4$,

$$R^2 = \frac{1}{4}J_4^2 - J_4 + I_4 = I_4.$$

Symmetry is immediate. Also $R\mathbf{1} = \mathbf{1}$, whereas $J_4 v = 0$ and hence $Rv = -v$ for $v \in \mathbf{1}^\perp$. □

Theorem 2.3 (Geometric Regge input)**.** *An elementary Regge move fixes θ_{01}, θ_{23} and replaces $\boldsymbol{\theta}$ by $R\boldsymbol{\theta}$. The same matrix acts on the corresponding four edge lengths, and the two tetrahedra have equal volume.*

Proof. This is the constant-curvature Regge theorem of Akopyan and Izmestiev [AI19, Theorem 1]. Only the angle statement is needed in the present section. □

2.2. **Phase coordinates and the meaning of "monomial".** For a physical tetrahedron define its six angle phases by

$$\begin{aligned} p &= e^{i\theta_{01}}, \quad a = e^{i\theta_{02}}, \quad b = e^{i\theta_{12}}, \\ q &= e^{i\theta_{23}}, \quad c = e^{i\theta_{13}}, \quad d = e^{i\theta_{03}}, \end{aligned} \qquad \tau = e^{iS}. \tag{2.9}$$

Then

$$\tau^2 = abcd. \tag{2.10}$$

Exponentiating (2.8) gives

$$(p, a, b, q, c, d, \tau) \longmapsto \left(p, \frac{\tau}{a}, \frac{\tau}{b}, q, \frac{\tau}{c}, \frac{\tau}{d}, \tau\right). \tag{2.11}$$

This is the exact meaning of the phrase "the Regge transformation acts monomially." A Laurent monomial in variables $z_1, \dots, z_m$ is a product $z_1^{n_1} \cdots z_m^{n_m}$ with integral exponents. Every new coordinate in (2.11) is such a Laurent monomial. The additive formula $S - \theta$ has become the multiplicative formula τ/a. Thus no choice of logarithm occurs in the algebraic formula.

To make that observation scheme-theoretic, set

$$\mathcal{X} = \operatorname{Spec} \frac{\mathbf{Q}[p^{\pm 1}, a^{\pm 1}, b^{\pm 1}, q^{\pm 1}, c^{\pm 1}, d^{\pm 1}, \tau^{\pm 1}]}{(\tau^2 - abcd)}. \tag{2.12}$$

Here $\operatorname{Spec} R$ means the affine algebraic space whose regular functions form the ring R, and $\mathbf{G}_m = \operatorname{Spec} \mathbb{Q}[u, u^{-1}]$ is the affine line with the point $u = 0$

removed. Thus an F-point of $\mathcal{X}$, for a field $F \supset \mathbb{Q}$, is simply a tuple of nonzero elements of F satisfying $\tau^2 = abcd$. This notation packages polynomial identities and changes of coordinates; no extra geometric hypothesis is hidden in it. Eliminating $d = \tau^2/(abc)$ identifies $\mathcal{X}$ with the six-dimensional algebraic torus $\mathbf{G}_m^6$. Physical phase data form a real-analytic subset of $\mathcal{X}(\mathbf{C})$, characterized in part by $|p| = |a| = \cdots = |q| = 1$. We enlarge to all complex points because $\mathcal{X}$, unlike the physical locus, is an algebraic parameter space defined over $\mathbf{Q}$.

Lemma 2.4 (Algebraic Regge involution)**.** *Formula* (2.11) *defines an involutive automorphism* $\mathcal{R} : \mathcal{X} \to \mathcal{X}$.

Proof. The product of the four transformed moving phases is

$$\frac{\tau^4}{abcd} = \tau^2,$$

so the defining relation of $\mathcal{X}$ is preserved. Applying the map twice sends, for example, a to $\tau/(\tau/a) = a$, and similarly for the other coordinates. The fixed coordinates remain fixed. □

The phase $u = e^{i\theta}$ recovers the corresponding Gram entry without choosing θ:

$$-\cos\theta = -\frac{u + u^{-1}}{2}. \tag{2.13}$$

Accordingly put

$$\begin{aligned} x &= -\frac{p + p^{-1}}{2}, & A &= -\frac{a + a^{-1}}{2}, & B &= -\frac{b + b^{-1}}{2}, \\ y &= -\frac{q + q^{-1}}{2}, & C &= -\frac{c + c^{-1}}{2}, & D &= -\frac{d + d^{-1}}{2}. \end{aligned} \tag{2.14}$$

In the order fixed above, the universal angle Gram matrix is

$$G = \begin{pmatrix} 1 & x & A & D \\ x & 1 & B & C \\ A & B & 1 & y \\ D & C & y & 1 \end{pmatrix}. \tag{2.15}$$

At a physical point this specializes to (2.2).

2.3. **The determinant orientation cover.** On the open subset of $\mathcal{X}$ where $\det G \neq 0$, adjoin a variable w satisfying

$$w^2 = \det G. \tag{2.16}$$

Because the characteristic is zero and $w \neq 0$, this is a finite étale double cover. Its deck involution is $w \mapsto -w$. A finite étale map is the algebraic analogue of a finite covering without branch points. Here this is elementary: the defining polynomial is $w^2 - \det G$, and its derivative $2w$ is invertible on the chosen open, so its two roots never collide.

The terminology "orientation cover" refers to the projective quadric which will be attached to G later. A smooth quadric surface over an algebraically

closed field has two families of maximal isotropic lines. For a four-dimensional quadratic form, choosing a square root of its determinant distinguishes these two families; changing the sign of the root interchanges them. Thus w, rather than the matrix G alone, retains the orientation needed by the quadric-scissors construction. At a physical spherical point we use the positive real root $w = \sqrt{\det G}$.

The determinant is Regge invariant. We now prove this in a form which also rationalizes the orientation cover.

Keep a, b, c, τ, q as independent invertible variables and eliminate

$$d = \frac{\tau^2}{abc}. \tag{2.17}$$

We temporarily forget the phase p, retaining only its cosine coordinate x. Define

$$s = \frac{q - q^{-1}}{2}, \qquad y = -\frac{q + q^{-1}}{2}. \tag{2.18}$$

Then

$$1 - y^2 = -s^2. \tag{2.19}$$

The asymmetry between p and q is deliberate: we descend p to x, while the retained phase q supplies the square root s in (2.19).

Define

$$\mathcal{B} = 2\{AB + DC - y(AC + DB)\}, \tag{2.20}$$

$$\mathcal{C} = 1 - A^2 - B^2 - C^2 - D^2 - y^2 + 2y(AD + BC) + (AC - BD)^2. \tag{2.21}$$

Lemma 2.5 (Determinant quadratic and Regge invariance)**.** *For the matrix* (2.15)*,*

$$\det G = s^2 x^2 + \mathcal{B}x + \mathcal{C}. \tag{2.22}$$

Under the monomial Regge substitution

$$(a, b, c, d) \longmapsto \left(\frac{\tau}{a}, \frac{\tau}{b}, \frac{\tau}{c}, \frac{\tau}{d}\right), \tag{2.23}$$

with p, q, τ fixed, one has

$$\mathcal{B}^R = \mathcal{B}, \qquad \mathcal{C}^R = \mathcal{C}. \tag{2.24}$$

Consequently $\det G^R = \det G$*, and the Regge involution has the lift*

$$(G, w) \longmapsto (G^R, w) \tag{2.25}$$

which does not change the chosen determinant root.

Proof. Expand the determinant of (2.15) as a polynomial in x. Its quadratic coefficient is $-(1-y^2) = s^2$, its linear coefficient is $2\{AB+DC-y(AC+DB)\}$, and its constant coefficient is the expression in (2.21). This proves (2.22).

For the invariance assertion, put

$$h(u) = \frac{u + u^{-1}}{2}$$

and introduce

$$I_1 = AB + CD, \qquad I_2 = AC + BD, \quad I_3 = AD + BC,$$
$$K = A^2 + B^2 + C^2 + D^2 + 4ABCD.$$

The elementary product-to-sum identity

$$2h(u)h(v) = h(uv) + h(u/v)$$

is an identity of Laurent polynomials. Under $a' = \tau/a, \dots, d' = \tau/d$, the relation $abcd = \tau^2$ gives

$$a'b' = cd, \quad c'd' = ab, \quad a'/b' = b/a, \quad c'/d' = d/c.$$

Since $h(u^{-1}) = h(u)$, the product-to-sum identity shows that $I_1^R = I_1$. Applying the same argument to the partitions $ac \mid bd$ and $ad \mid bc$ gives $I_2^R = I_2$ and $I_3^R = I_3$.

It remains to treat K. Set

$$P = h(ab), \quad Q = h(cd), \quad r = h(a/b), \quad s_0 = h(c/d).$$

Direct applications of the same product-to-sum identity give

$$h(a)^2 + h(b)^2 = 1 + Pr,$$
$$h(c)^2 + h(d)^2 = 1 + Qs_0,$$
$$4h(a)h(b)h(c)h(d) = (P + r)(Q + s_0).$$

Because $A = -h(a), \dots, D = -h(d)$, it follows that

$$K = 2 + PQ + rs_0 + (P + Q)(r + s_0).$$

The Regge substitution swaps P, Q and fixes r, s_0, so $K^R = K$. Finally,

$$\frac{\mathcal{B}}{2} = I_1 - yI_2, \qquad \mathcal{C} = 1 - y^2 + 2yI_3 + I_2^2 - K,$$

where the second equality uses $(AC - BD)^2 = (AC + BD)^2 - 4ABCD$. Thus $\mathcal{B}^R = \mathcal{B}$ and $\mathcal{C}^R = \mathcal{C}$ inside the Laurent coordinate ring of $\mathcal{X}$, not merely on the physical locus. Since x, y, s are fixed, the determinant quadratics coincide. Keeping the same root w therefore gives the displayed lift. □

2.4. **Rational charts on the orientation cover.** Let

$$S_0 = \mathbf{Q}[a^{\pm 1}, b^{\pm 1}, c^{\pm 1}, \tau^{\pm 1}, q^{\pm 1}],$$

and regard $d, A, B, C, D, y, s, \mathcal{B}, \mathcal{C}$ as the elements of S_0 defined above. The descended orientation space is the affine hypersurface

$$\mathcal{Y} = \operatorname{Spec} \frac{S_0[x, w]}{(w^2 - s^2x^2 - \mathcal{B}x - \mathcal{C})}. \tag{2.26}$$

The orientation cover of the phase torus maps to $\mathcal{Y}$ by $x = -(p + p^{-1})/2$. The point of passing to $\mathcal{Y}$ is that it is rational, although the original equation visibly contains a square root.

Proposition 2.6 (Two rational orientation charts)**.** *Put*

$$r = w - sx, \qquad r_+ = w + sx. \tag{2.27}$$

On the principal open set where $2sr - \mathcal{B} \neq 0$*, one has*

$$x = \frac{\mathcal{C} - r^2}{2sr - \mathcal{B}}, \qquad w = r + sx. \tag{2.28}$$

On the principal open set where $-2sr_+ - \mathcal{B} \neq 0$*, one has*

$$x = \frac{\mathcal{C} - r_+^2}{-2sr_+ - \mathcal{B}}, \qquad w = r_+ - sx. \tag{2.29}$$

Each formula identifies the indicated part of $\mathcal{Y}$ *with an open subset of* $\mathbf{G}_m^5 \times \mathbf{A}^1$*. In particular, its function field is purely transcendental:*

$$\mathbf{Q}(a, b, c, \tau, q, r) \quad \text{or} \quad \mathbf{Q}(a, b, c, \tau, q, r_+). \tag{2.30}$$

On the locus $sw \neq 0$*, the two charts cover* $\mathcal{Y}$*.*

Proof. Substitute $w = r + sx$ into (2.22):

$$r^2 + 2rsx + s^2x^2 = s^2x^2 + \mathcal{B}x + \mathcal{C}.$$

After cancellation of the quadratic terms,

$$(2sr - \mathcal{B})x = \mathcal{C} - r^2,$$

which is (2.28). Conversely, the two formulas in (2.28) satisfy both $r = w - sx$ and the defining equation of $\mathcal{Y}$; hence they give mutually inverse coordinate maps on the stated principal opens. The proof of (2.29) is identical after substituting $w = r_+ - sx$.

It remains to check coverage. Written in the original coordinates, the two denominators are

$$2s(w - sx) - \mathcal{B}, \qquad -2s(w + sx) - \mathcal{B}.$$

If both vanished, subtracting the second equation from the first would give $4sw = 0$. This is impossible on the locus $sw \neq 0$. □

The change of variable has a simple geometric-algebraic interpretation. The equation $w^2 = s^2x^2 + \mathcal{B}x + \mathcal{C}$ is a conic in the (x, w)-plane over the field $\mathbf{Q}(a, b, c, \tau, q)$. The coordinates $r = w - sx$ and $r_+ = w + sx$ subtract one or the other asymptotic linear part. This cancels the x^2-term and leaves an equation linear in x. No continuity or density argument is involved.

By Lemma 2.5, the Regge involution fixes $x, w, q, s, \mathcal{B}, \mathcal{C}$. It therefore fixes r and r_+ and preserves both rational charts. The two oriented Gram matrices (G, w) and (G^R, w) are defined over the same chart.

2.5. **The universal open set.** The rational chart contains points which do not represent nondegenerate tetrahedra, and it also contains matrices with degenerate coordinate restrictions. These unwanted points lie on algebraic divisors and may be removed once and for all.

Fix one of the two charts in Proposition 2.6.

For a regular function f, the *principal open* $D(f)$ is obtained by removing its zero set, equivalently by adjoining f^{-1} to the coordinate ring. The *function field* of an irreducible affine variety is the field of fractions of its coordinate ring. Shrinking to a nonempty principal open changes the ring of regular functions but not this field of rational functions.

Definition 2.7 (Universal open)**.** Let U be the Regge-stable principal open obtained by requiring:

(1) the displayed chart denominator, s, w, and $1 - x^2$ are nonzero;
(2) G and G^R are nonsingular; and
(3) every proper principal restriction of G and G^R which occurs as a coordinate face is nonsingular.

Equivalently, take the product of these finitely many functions and their Regge transforms and invert that product. This makes the definition manifestly Regge stable. Subsequent constructions may shrink U further by finitely many Regge-stable nonvanishing conditions; such a shrinking does not change its function field.

Since a rational chart is an open subset of $\mathbf{G}_m^5 \times \mathbf{A}^1$, every nonempty such U is smooth and irreducible, and

$$\mathbf{Q}(U) = \mathbf{Q}(a, b, c, \tau, q, r) \quad \text{or} \quad \mathbf{Q}(a, b, c, \tau, q, r_+). \tag{2.31}$$

The word "universal" does not mean that every complex point of U is a convex spherical tetrahedron. It means that G, G^R, their common orientation root, and the Regge involution are all defined simultaneously over one algebraic base. The physical spherical pairs form a distinguished real locus inside its complex points.

Proposition 2.8 (Every physical pair lies on a rational chart)**.** *Let $T, T^R \subset S^3$ be a nondegenerate spherical Regge pair. Choose the physical phases (2.9) and the positive root $w = \sqrt{\det G(T)}$. Then the resulting oriented point belongs to at least one of the two rational charts. After shrinking that chart as above, it is a complex point of a universal open U, and its Regge mate is the image of that point under $\mathcal{R}$.*

Proof. Because $0 < \theta_{23} < \pi$,

$$s = \frac{e^{i\theta_{23}} - e^{-i\theta_{23}}}{2} = i \sin \theta_{23} \neq 0.$$

Nondegeneracy and positive definiteness give $w > 0$. Thus $sw \neq 0$, and Proposition 2.6 puts the point on at least one chart. All principal restrictions of a positive-definite matrix are positive definite; the same applies to the

nondegenerate spherical mate. Hence none of the functions removed in the definition of U vanishes at the pair. The common-root lift (2.25) identifies the second oriented matrix with the Regge image of the first. □

This is the step which removes any algebraicity restriction on an individual tetrahedron. Its angles may be transcendental: they merely specify a complex point of the algebraic family U. The later argument will prove one identity over U and then pull that identity back to this point.

2.6. **Recovering the omitted fixed phase.** The functions $a, b, c, q, d = \tau^2/(abc)$ are already defined on U. The only original angle phase which is not a function on U is p, because the rational chart remembers only

$$x = -\frac{p + p^{-1}}{2}.$$

Equivalently, p satisfies

$$p^2 + 2xp + 1 = 0. \tag{2.32}$$

Proposition 2.9 (The fixed-phase cover)**.** *Define*

$$\pi : U_p = \operatorname{Spec} \frac{\mathcal{O}_U[p]}{(p^2 + 2xp + 1)} \longrightarrow U. \tag{2.33}$$

Then π is finite étale of degree two. Its nontrivial deck transformation is

$$p \longmapsto p^{-1} = -2x - p. \tag{2.34}$$

The cover is common to G and G^R, because the Regge transformation fixes the 01*-angle and hence fixes x.*

Proof. The polynomial in (2.32) is monic of degree two, so U_p is finite locally free of rank two over U. Its discriminant is

$$(2x)^2 - 4 = 4(x^2 - 1). \tag{2.35}$$

The function $1 - x^2$ was inverted in the definition of U; hence the discriminant is a unit and the cover is étale. The product of the two roots is one, so the second root is p^{-1}, and their sum is $-2x$, giving (2.34). Finally, x is fixed by the Regge involution. □

On a physical fibre the two points of U_p are $e^{i\theta_{01}}$ and $e^{-i\theta_{01}}$. The cover may split after base change, but it need not split over U itself. This two-sheeted fixed-phase datum is the sole residual effect of descending from the full phase torus to the rational orientation chart.

3. The motivic dictionary and the exact interface

This section explains the intended geometry behind Inputs (E2)–(E5) and then restates those inputs as ordinary linear algebra. It is intentionally not presented as a construction of a relative category of motives. A reader interested only in the logical proof may start at Subsection 3.5.

3.1. **From a spherical simplex to an oriented quadric simplex.** Let U be one of the smooth rational opens of Section 2. Over it we have a rank-four trivial bundle and two symmetric invertible matrices G, G^R. The equation

$$Q_G : \quad z^t G z = 0$$

defines a relative quadric in $\mathbf{P}^3_U$; the four coordinate hyperplanes $M_i = \{z_i = 0\}$ are its ordered faces.

The spherical tetrahedron first produces $Q_{G^{-1}}$. Projective duality replaces it by Q_G: if $x = Gy$, then

$$x^t G^{-1} x = y^t G y, \qquad e_j^t G^{-1} x = y_j.$$

Thus the polar faces become coordinate faces. This elementary change of coordinates explains why Section 4 calculates with G, although Goncharov's spherical generator is written using G^{-1}. Input (E6) includes the assertion that projective duality is an involution on the relevant quadric-scissors group.

A smooth quadric surface over an algebraically closed field has two families of maximal isotropic lines, called its two rulings. An orientation is a choice between them. For a matrix quadric, a square root $w^2 = \det G$ distinguishes the choices; changing w interchanges them. This is why the parameter space in Section 2 is the determinant double cover.

3.2. **What a motive is.** Ordinary cohomology sends a space directly to groups or vector spaces. A motive is intended to be a universal algebraic object from which several cohomology theories—Betti, de Rham, Hodge, or étale, for example—are obtained by realization functors. One keeps the object itself because the proof needs operations which a list of Betti numbers forgets: pullback in a family, tensor products, duality, Tate twists, localization triangles, filtrations, and extension classes.

The customary ambient notation is

$$\mathrm{DM}(U)_{\mathbb{Q}}.$$

Morally, this is a rational, symmetric monoidal, stable category generated by smooth U-schemes and localization. The unit is $\mathbb{Q}_U(0)$; the Tate object is $\mathbb{Q}_U(1)$; shifts are denoted by $[m]$. In such a category motivic cohomology is the Hom group

$$H^m(U, \mathbb{Q}(n)) = \mathrm{Hom}_{\mathrm{DM}(U)_{\mathbb{Q}}}\big(\mathbb{Q}_U(0), \mathbb{Q}_U(n)[m]\big). \tag{3.1}$$

This formula explains the notation used later. It does not, by itself, construct the category or prove any comparison theorem.

Why not replace every motive by a complex of rational vector spaces? That replacement destroys the obstruction on which the proof depends. The category of rational vector spaces is semisimple, so

$$\mathrm{Ext}^1_{\mathrm{Vect}_{\mathbb{Q}}}(\mathbb{Q}, \mathbb{Q}) = 0,$$

and every bounded complex is quasi-isomorphic to the direct sum of its cohomology groups. A nontrivial motivic extension

$$0 \longrightarrow \mathbb{Q}(2) \longrightarrow E \longrightarrow \mathbb{Q}(0) \longrightarrow 0$$

would therefore become split after forgetting to plain vector spaces. The lost extension is precisely the dilogarithmic or K_3 information that distinguishes a primitive weight-two class. Motives are not ornate terminology for ordinary homology; they retain the extension data which ordinary rational homology cannot see.

The conventions relevant here are

$$H^1(W, \mathbb{Q}(1)) \cong \mathcal{O}(W)^* \otimes_{\mathbf{Z}} \mathbb{Q}, \qquad H^1(F, \mathbb{Q}(2)) \cong \operatorname{gr}_2^\gamma K_3(F) \otimes \mathbb{Q}$$

for the schemes and fields to which the external comparison theorems apply. The first formula turns cross-ratios into weight-one classes. The second explains why the endpoint obstruction belongs to K_3.

Here $K_3(F)$ is Quillen's third algebraic K-group: it is the third homotopy group of the algebraic K-theory space built from invertible matrices over F. Exterior-power operations give its gamma filtration; $\operatorname{gr}_2^\gamma K_3(F)$ is the second associated quotient of that filtration. No presentation or computation of the full group is needed in this proof. The notation $\operatorname{Ext}^1(X, Y)$ means equivalence classes of short exact sequences

$$0 \longrightarrow Y \longrightarrow E \longrightarrow X \longrightarrow 0.$$

Thus the proposed class in $H^1(U, \mathbb{Q}(2))$ records a possibly nonsplit extension between the two endpoint Tate objects.

3.3. **Tate and sign-Artin objects.** The symbol $\mathbb{Q}_U(n)$ denotes the n-fold Tate twist. The indices $0, 1, 2$ below are Tate degrees; they are not homological degrees. A Tate object is the basic one-dimensional motivic object corresponding in degree one to the multiplicative group; tensoring by it shifts Tate degree. An Artin object is the motive of a finite étale scheme and, over a field, behaves like a rational representation of its finite Galois group. A quadratic finite étale cover $\pi : U_p \to U$, with deck involution σ, is expected to give

$$\pi_* \mathbb{Q}_{U_p} \cong \mathbb{Q}_U \oplus \chi, \qquad \chi = \operatorname{im} \frac{1 - \sigma}{2}. \tag{3.2}$$

The summand χ is the sign-Artin object. If the cover splits, this summand is another copy of $\mathbb{Q}_U$. Abstractly the projector exists because 2 is invertible over $\mathbb{Q}$. The assertion that this decomposition, pullback, transfer, and the relevant framed coefficient spaces are all compatible is the Artin-trace clause of Input (E4).

3.4. **The intended four-term face object.** For $I \subset \{0, 1, 2, 3\}$, put

$$M_I = \bigcap_{i \in I} M_i, \qquad Q_I = Q \cap M_I.$$

Ordering the hyperplanes gives alternating incidence signs. The geometric expression behind the relative coefficient is

$$\begin{aligned}\mathcal{F}_U(Q,M):\quad \bigoplus_{|I|=3} M_U(M_I\setminus Q_I) &\longrightarrow \bigoplus_{|I|=2} M_U(M_I\setminus Q_I)\\ &\longrightarrow \bigoplus_{|I|=1} M_U(M_I\setminus Q_I)\longrightarrow M_U(\mathbf{P}^3_U\setminus Q).\end{aligned} \tag{3.3}$$

Here $M_U(X)$ denotes the proposed geometric motive of a smooth U-scheme X. Totalization is the stable-category analogue of forming one object from a bounded chain complex; it may be described by successive mapping cones. The internal Hom into $\mathbb{Q}_U$ is its cohomological dual, and the final (2) is the normalization which places the endpoint Tate degrees at 0 and 2. If a relative motivic category with functorial totalizations and duals has been fixed, the intended normalized object is

Definition 3.1 (Intended normalized relative quadric motive)**.**

$$\mathfrak{m}_U(Q,M)=\underline{\mathrm{Hom}}\big(\mathrm{Tot}\,\mathcal{F}_U(Q,M),\mathbb{Q}_U\big)(2). \tag{3.4}$$

Definition 3.1 fixes the signs, dual, and Tate twist which Inputs (E2)–(E5) must use. It is a specification, not a claim that the indicated totalization has been constructed in Lean or that all of its relative comparison properties follow formally from a citation. Its field-level face-complex convention is the one used by Goncharov [Gon99]; the twist agrees with Brown's normalization [Bro13, Definition 5.8]. The Aomoto-pair construction of Beilinson–Goncharov–Schechtman–Varchenko is a historical precursor [BGSV90]. None of these citations is being used as an unstated proof of the relative base-change assertions in Input (E2).

The expected localization calculation has three graded pieces:

$$\mathrm{gr}^0\,\mathfrak{m}_U\cong\mathbb{Q}_U(0),\qquad \mathrm{gr}^1\,\mathfrak{m}_U\cong\bigoplus_{|I|=2}\widetilde{M}(Q_I)(1),\qquad \mathrm{gr}^2\,\mathfrak{m}_U\cong\mathbb{Q}_U(2). \tag{3.5}$$

Here $\widetilde{M}(Q_I)$ is the reduced degree-two Artin object of the binary section. In the Regge family five such objects are split and the sixth is either split or χ. The orientation supplies a frame on one endpoint and the ordered faces supply a frame on the other. The existence of a strict heart containing these objects, the filtration (3.5), and its compatibility with the frames are exactly the non-formal part of Input (E3).

A t-structure on a stable category selects an abelian *heart* in which appropriate triangles become short exact sequences. A three-weight heart here would be the extension closure of the three kinds of pure object in Equation (3.5). A framing chooses the generator of the top $\mathbb{Q}(0)$-piece and the covector on the bottom $\mathbb{Q}(2)$-piece. The reduced coproduct of the resulting matrix coefficient records all factorizations from the top frame to the bottom frame through the middle degree. Consequently a zero reduced coproduct is

called *primitive*: no middle factorization remains, so the only possible datum is the direct endpoint extension. Input (E3) asserts that this description is valid for the proposed relative objects.

3.5. **The linear contract used by the proof.** Nothing after this point requires manipulating a category of motives. The conditional proof uses the following data.

Definition 3.2 (Weight-two coefficient contract)**.** For a chart U, a weight-two coefficient contract consists of rational vector spaces

$$A_{02}(U), \qquad A_{11}(U), \qquad H_U,$$

a linear map

$$\overline{\Delta}_U : A_{02}(U) \longrightarrow A_{11}(U),$$

an isomorphism

$$\mathrm{prim}_U : \ker \overline{\Delta}_U \xrightarrow{\sim} H_U, \tag{3.6}$$

and a proof that $H_U = 0$. It also contains the two framed coefficients

$$c_U(G), \qquad c_U(G^R),$$

the six-channel coproduct formula of Input (E4), a linear specialization map on the middle coproduct which commutes with the field-level quadric coproduct, and the linear coefficient-specialization and endpoint maps of Input (E5). The field-level map c_G is typed only on the kernel of that field-level coproduct; primitivity is therefore a proof obligation, not a suggestive name attached to an arbitrary subgroup.

The intended interpretation is

$$H_U = \mathrm{Ext}^1\big(\mathbb{Q}_U(0), \mathbb{Q}_U(2)\big) = H^1(U, \mathbb{Q}(2)).$$

Calling an element primitive means only that its reduced coproduct is zero. The following deduction is elementary and is the part formalized in Lean.

Lemma 3.3 (Primitive vanishing)**.** *In a weight-two coefficient contract, if* $\overline{\Delta}_U(x) = 0$*, then* $x = 0$*.*

Proof. The equation lets us regard x as an element $\widetilde{x} \in \ker \overline{\Delta}_U$. Its image under prim_U lies in the zero vector space H_U, hence is zero. Since prim_U is injective, $\widetilde{x} = 0$, and therefore $x = 0$. □

This lemma is intentionally weaker than constructing a motivic heart. The deep assertion is that the geometric coefficient contract exists and that prim_U is the motivic primitive–extension identification. That assertion is Input (E3).

3.6. **Why the proposed obstruction vanishes.** The suggested mathematical route to the last field of the contract is

$$\begin{aligned} H_U &= H^1(U, \mathbb{Q}(2)) \\ &\cong H^1(\mathbb{Q}(t_1, \dots, t_6), \mathbb{Q}(2)) \\ &\cong H^1(\mathbb{Q}, \mathbb{Q}(2)) \\ &\cong \mathrm{gr}_2^\gamma K_3(\mathbb{Q}) \otimes \mathbb{Q} = 0. \end{aligned} \tag{3.7}$$

The middle comparisons are the intended application of Garkusha's birational and pure-transcendental results [Gar23]; the final vanishing uses Borel's rank theorem [Bor77]. For the conditional proof, Equation (3.7) is not rederived inside an unspecified category $\mathrm{DM}(U)_{\mathbb{Q}}$. Its exact applicability to the chosen open and to the heart-to-derived comparison is bundled into Input (E3).

3.7. **Specialization and the boundary of the interface.** At a complex point $s : \operatorname{Spec} \mathbb{C} \to U$, the intended motivic construction would provide a pullback

$$s^* \mathfrak{m}_U(Q, M) \cong \mathfrak{m}_{\mathbb{C}}(Q_s, M_s)$$

preserving totalization, internal duality, twist, both frames, the three-step filtration, and the endpoint map. Goncharov's construction over $\mathbb{C}$ would then identify that endpoint with c_G of the corresponding primitive quadric class.

This is exactly Input (E5). The manuscript does not infer it from an unspecified appeal to functoriality. In particular, the following five assertions remain on the external side of the proof:

(i) construction of the required relative $\mathrm{DM}(U)_{\mathbb{Q}}$ objects and their functorial totalizations;
(ii) the relative localization calculation and strict three-level filtration;
(iii) primitive coefficients $= \mathrm{Ext}^1 = H^1(U, \mathbb{Q}(2))$;
(iv) twisted-trace compatibility for the sign-Artin channel; and
(v) coproduct-compatible specialization, followed by equality of the specialized endpoint, with all signs and twists, to Goncharov's field-level map c_G.

The rest of the manuscript never uses a stronger motivic statement than these five clauses.

4. The universal defect and its coproduct

Fix a physical point on one of the two charts supplied by Proposition 2.8. Shrink its connected principal open U, without removing that point, so that all matrices, cross-ratios, and finite root covers mentioned below are regular. This does not change the function field $\mathbb{Q}(t_1, \dots, t_6)$ of the chart.

4.1. **The matrix geometry behind the motivic input.** Let G and G^R be the universal angle Gram matrices with their common determinant root w. We work with the projectively dual coordinate quadrics

$$Q_G : \ z^t G z = 0, \qquad Q_{G^R} : \ z^t G^R z = 0.$$

The following calculation explains the shape of the three-weight package. It does not construct that package.

Lemma 4.1 (Coordinate-stratum calculation)**.** *At the level of quadratic forms, Q_G and Q_{G^R} are hyperbolic on U; every coordinate rank-three restriction has an isotropic line. Five coordinate binary restrictions have their two isotropic points displayed by the phases a, b, c, d, q. The sixth has isotropic points over the common quadratic cover*

$$U_p = U[p]/(p^2 + 2xp + 1),$$

whose deck transformation is $p \mapsto p^{-1}$.

Proof. For a unit u, put $g = -(u + u^{-1})/2$. Then $(1, u)$ is isotropic for

$$\begin{pmatrix} 1 & g \\ g & 1 \end{pmatrix}, \qquad 1 + 2gu + u^2 = 0.$$

In particular, the q-block is a hyperbolic plane. Its binary Schur complement C in G satisfies

$$\det C = \frac{\det G}{1 - y^2} = -\left(\frac{w}{s}\right)^2, \qquad s = \frac{q - q^{-1}}{2}.$$

Put $t = w/s$, $J_0 = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$, and $J = t^{-1} J_0 C$. The identities

$$(J_0 C)^2 = -\det(C) I, \qquad J^2 = I, \qquad J^t C + C J = 0$$

show that the two eigenlines of J are isotropic and paired perfectly by C. Thus the complementary plane, and hence the rank-four form, is hyperbolic. The same q, s, w are fixed by Regge, so the calculation also applies to G^R.

Every coordinate rank-three restriction contains one of the displayed binary blocks and hence an isotropic vector. The five phase variables give the two roots of five binary restrictions. The remaining restriction has equation $p + p^{-1} = -2x$, equivalently $p^2 + 2xp + 1 = 0$, and Proposition 2.9 gives its deck transformation. □

Passing from this calculation to relative motives of all strata, including regularity, functorial localization, frames, and the reduced sign-Artin summand, is part of Inputs (E2)–(E4). We do not relabel that passage as a proof by localization.

4.2. **The relative coefficient supplied by the contract.** Apply Input (E2) to the two oriented families. In the notation of Definition 3.2, set

$$c_G = c_U(G), \qquad c_R = c_U(G^R), \qquad \delta_U^\vee = c_G - c_R \in A_{02}(U).$$

The superscript $\vee$ records projective duality $Q_{G^{-1}} \leftrightarrow Q_G$; it is not the internal dual used in the intended face motive.

Input (E3) supplies the following statement. It is displayed as an external proposition so that its logical status cannot be mistaken.

Proposition 4.2 (External three-weight package)**.** *The two coefficients have endpoint types* $(0,2)$*, their middle terms are five Tate channels and one possible sign-Artin channel, and*

$$\mathrm{prim}_U : \ker \overline{\Delta}_U \xrightarrow{\sim} H^1(U, \mathbb{Q}(2))$$

is an isomorphism. Moreover $H^1(U, \mathbb{Q}(2)) = 0$*.*

This proposition is Input (E3), not a theorem proved from the coordinate-stratum calculation. Its proposed motivic graded pieces are

$$\mathbb{Q}_U(0), \qquad \mathbb{Q}_U(1)^{\oplus 5} \oplus \chi(1), \qquad \mathbb{Q}_U(2),$$

and its proposed vanishing route is Equation (3.7).

4.3. **Length–angle channels.** Input (E4) identifies the reduced coproduct of a dual oriented quadric coefficient with

$$\overline{\Delta}_U c_G = \sum_e L_e \otimes A_e, \tag{4.1}$$

one term for each physical edge. The angle factor comes from a coordinate binary section and the length factor from its complementary binary Schur form. This ordering is important: it is $L_e \otimes A_e$.

Here is the elementary cross-ratio normalization. For

$$r(z_1, z_2, z_3, z_4) = \frac{(z_1 - z_3)(z_2 - z_4)}{(z_1 - z_4)(z_2 - z_3)}$$

and two real rays of angle ϕ, use slopes $0, \tan\phi$ and order the isotropic directions as $-\mathrm{i}, \mathrm{i}$. Direct substitution gives

$$r(-\mathrm{i}, \mathrm{i}, 0, \tan\phi) = \frac{1 + \mathrm{i}\tan\phi}{1 - \mathrm{i}\tan\phi} = e^{2\mathrm{i}\phi}.$$

Thus, with compatible orientations,

$$A_e = e^{2\mathrm{i}\vartheta_e}, \qquad L_e = e^{2\mathrm{i}\ell_e}. \tag{4.2}$$

Swapping the two isotropic directions inverts the cross-ratio. Input (E4) asserts that its relative formula uses exactly these orientations and that the same convention survives specialization.

4.4. **The internal tensor calculation.** Let

$$R = \frac{1}{2}J_4 - I_4, \qquad H = 2R = J_4 - 2I_4.$$

Section 2 proves $R^tR = I_4$, hence

$$H^tH = 4I_4. \tag{4.3}$$

For the four moving edges, choose half-phases

$$A_i = B_i^2, \qquad L_i = Y_i^2.$$

The universalization clause of Input (E4) says that, on a common finite splitting cover, the real Regge relations of Input (E1) extend to the algebraic identities

$$A_i^R = \prod_j B_j^{H_{ij}}, \qquad L_i^R = \prod_j Y_j^{H_{ij}}. \tag{4.4}$$

This clause includes the Zariski-density argument and the algebraic recovery of length phases from cofactors; neither is silently inferred from the formal matrix identity.

Lemma 4.3 (Regge tensor cancellation)**.** *Let $B_1, \dots, B_4$ and $Y_1, \dots, Y_4$ lie in abelian multiplicative groups, tensor those groups over $\mathbf{Z}$, and then tensor with $\mathbb{Q}$. If $A_i = B_i^2$, $L_i = Y_i^2$, and the transformed terms obey (4.4), then*

$$\sum_i L_i^R \otimes A_i^R = \sum_i L_i \otimes A_i.$$

Proof. Write a multiplicative group additively inside its tensor product. Then $[u^m] = m[u]$, so Equation (4.4) and $H^tH = 4I_4$ give

$$\begin{aligned}\sum_i L_i^R \otimes A_i^R &= \sum_{i,j,k} H_{ik}H_{ij}\,[Y_k] \otimes [B_j] \\ &= 4\sum_j [Y_j] \otimes [B_j] \\ &= \sum_j [Y_j^2] \otimes [B_j^2] = \sum_j L_j \otimes A_j.\end{aligned}$$

Changing a square root changes its class by two-torsion, which disappears after tensoring with $\mathbb{Q}$. □

This is the entire Regge-specific coproduct computation. Lean proves both this half-phase form using H and the equivalent full-channel form using the normalized matrix $R = H/2$. The latter is the identity connected directly to the six-channel coproduct contract.

4.5. **Primitive and universal vanishing.**

Lemma 4.4 (Vanishing of the middle coproduct)**.** *Under Inputs* (E1), (E2), and (E4)*,*

$$\overline{\Delta}_U(\delta_U^\vee) = 0.$$

At every complex point $s : \operatorname{Spec}\mathbb{C} \to U$, the specialized dual quadric-scissors defect is primitive as well.

Proof. The two channels belonging to the selected opposite edge pair agree term-by-term because Regge fixes the corresponding lengths and angles. On the common splitting cover, the remaining four channels agree by Lemma 4.3. The descent clause of Input (E4) says that pullback is injective on the five Tate channels and, using normalized twisted trace, on the possible sign-Artin channel. Hence the equality descends to $A_{11}(U)$. The coproduct-specialization square in the same input carries this zero to Goncharov's field-level coproduct. Thus the specialized defect lies in its kernel; this is exactly the proof term which licenses applying c_G, rather than an independently assumed primitive label. □

Theorem 4.5 (Vanishing of the universal framed defect)**.** *Under Inputs* (E1)–(E4)*,*

$$\delta_U^\vee = 0 \qquad \text{in } A_{02}(U).$$

Proof. Lemma 4.4 makes the defect primitive. Proposition 4.2 supplies the weight-two coefficient contract. Lemma 3.3 therefore gives $\delta_U^\vee = 0$. □

The proof of this theorem is now transparently divided. Orthogonality and the tensor cancellation are internal and machine checked. Construction of the relative coefficients, the universalization and descent of their coproduct, and primitive = motivic extension are the named external clauses.

5. Specialization and the return to scissors congruence

We now prove Theorem 1.1 from the universal vanishing theorem and Inputs (E5)–(E8). All arrows in the argument are named before the diagram chase begins.

5.1. **The field-level comparison maps.** Let $S_4(\mathbb{C})_\mathbb{Q}$ be Goncharov's rational vector space of oriented quadric simplices: generators are smooth oriented quadrics in $\mathbf{P}^3$ with four ordered hyperplane faces, modulo projective, alternating, degeneracy, and additivity relations. Its weight-two differential is the reduced coproduct

$$\overline{\Delta}_\mathbb{C} : S_4(\mathbb{C})_\mathbb{Q} \longrightarrow S_2(\mathbb{C})_\mathbb{Q} \otimes_\mathbb{Q} S_2(\mathbb{C})_\mathbb{Q},$$

where $S_2(\mathbb{C})_\mathbb{Q} \cong \mathbb{C}^* \otimes_\mathbf{Z} \mathbb{Q}$. Since the normalized cobar complex has no preceding term in this degree,

$$H^1(S^\bullet(2))_\mathbb{Q} = \ker \overline{\Delta}_\mathbb{C}. \tag{5.1}$$

Theorem 5.1 (External: Goncharov's weight-two comparison)**.** *There is an injective linear map*

$$c_G : \ker \overline{\Delta}_{\mathbb{C}} \longrightarrow \mathrm{gr}_2^{\gamma} K_3(\mathbb{C}) \otimes \mathbb{Q}. \tag{5.2}$$

This is the case $n = 2, i = 1$ of Goncharov's Theorem 4.13 and equations (4.6)–(4.7) [Gon99]. It is Input (E6); the manuscript does not formalize algebraic K-theory or reprove injectivity.

Define the reduced spherical scissors group by

$$\widetilde{\mathcal{P}}(S^3) = \mathrm{coker}\big(\Sigma : \widehat{\mathcal{P}}(S^2) \longrightarrow \widehat{\mathcal{P}}(S^3)\big),$$

where Σ is spherical suspension. Geometrically, if a polygon lies in an equatorial S^2, its suspension is the double cone obtained by joining it to the two antipodal poles. Cutting makes this construction an additive map on scissors groups.

Theorem 5.2 (External: spherical presentation and quadric map)**.** *The reduced geometric group $\widetilde{\mathcal{P}}(S^3)$ is canonically the spherical configuration group used by Goncharov. After rationalization, complexification gives an injective map*

$$\psi_S : \widetilde{\mathcal{P}}(S^3) \otimes \mathbb{Q} \hookrightarrow S^{\mathrm{sph}}_{4,\mathbb{Q}}, \tag{5.3}$$

where $S^{\mathrm{sph}}_{4,\mathbb{Q}}$ is the conjugation-invariant spherical part of $S_4(\mathbb{C})_{\mathbb{Q}}$ with its one-dimensional parity line. The parity line records how reversing the chosen ruling changes an oriented quadric generator; tensoring with that line is what makes the map descend from the unoriented geometric scissors group fixed in Section 1. For a tetrahedron with angle Gram matrix G, its image is represented by $Q_{G^{-1}}$ with its ordered coordinate hyperplanes.

The presentation comparison, including its value on this tetrahedral generator and hence on the Regge difference, is Input (E7), cited from Dupont's Theorem 7.14 [Dup01]; the map and its injectivity are Input (E6), cited from Goncharov's Theorem 4.14 [Gon99]. We suppress the nonzero one-dimensional parity factor when testing whether a class vanishes.

Projective duality gives an involutive linear automorphism

$$D : S_4(\mathbb{C})_{\mathbb{Q}} \xrightarrow{\sim} S_4(\mathbb{C})_{\mathbb{Q}}$$

which takes the $Q_{G^{-1}}$-class to the coordinate Q_G-class. Input (E6) includes both this map and its compatibility with the two displayed Gram-matrix generators. Consequently $D(x) = 0$ if and only if $x = 0$.

5.2. **The family-to-fibre bridge.** The last comparison is not a formal consequence of merely having a pullback functor.

Proposition 5.3 (External: fibre endpoint comparison)**.** *Let $s : \mathrm{Spec}\,\mathbb{C} \to U$ be a complex point and let $\delta_s^{\vee} \in S_4(\mathbb{C})_{\mathbb{Q}}$ be the projectively dual quadric-scissors defect. If it is primitive, then the specialization of the endpoint of the relative coefficient defect is exactly*

$$\mathrm{endpoint}_s(\delta_U^{\vee}) = c_G(\delta_s^{\vee}). \tag{5.4}$$

The equation uses the same face-complex signs, ruling frame, ordered-face frame, projective duality, and Tate twist on both sides.

The specialization and endpoint arrows in this equation are linear maps. This proposition is Input (E5). Goncharov's face-motive construction over a field suggests it [Gon99], and Definition 3.1 fixes the candidate relative object. The exact relative base-change and normalization comparison is not proved in this manuscript and is not attributed to a field-level theorem.

5.3. **The machine-checkable diagram chase.** Let T, T^R be the pair in Theorem 1.1. By Proposition 2.8, their angle phases and positive determinant root define a point

$$s_T : \operatorname{Spec} \mathbb{C} \longrightarrow U.$$

Put

$$z = [T] - [T^R] \in \widehat{\mathcal{P}}(S^3), \qquad \bar{z} \in \widetilde{\mathcal{P}}(S^3),$$

and let

$$\delta_{s_T} = \psi_S(\bar{z} \otimes 1), \qquad \delta_{s_T}^\vee = D(\delta_{s_T}).$$

Proposition 5.4 (Reduced scissors vanishing)**.** *Under Inputs* (E1)–(E8)*,*

$$\bar{z} = 0 \qquad \text{in } \widetilde{\mathcal{P}}(S^3).$$

Proof. Lemma 4.4 gives $\overline{\Delta}_{\mathbb{C}}(\delta_{s_T}^\vee) = 0$, so $\delta_{s_T}^\vee$ lies in the domain of c_G. Theorem 4.5 gives $\delta_U^\vee = 0$. The endpoint map is linear, hence its value on this zero class is zero. Proposition 5.3 therefore gives

$$c_G(\delta_{s_T}^\vee) = 0.$$

Injectivity in Theorem 5.1 yields $\delta_{s_T}^\vee = 0$. Since D is an involution, $\delta_{s_T} = 0$. Injectivity of ψ_S gives

$$\bar{z} \otimes 1 = 0 \quad \text{in } \widetilde{\mathcal{P}}(S^3) \otimes \mathbb{Q}.$$

By Input (E8), $\widehat{\mathcal{P}}(S^2)$ and $\widehat{\mathcal{P}}(S^3)$ are $\mathbb{Q}$-vector spaces. The image of suspension and its cokernel $\widetilde{\mathcal{P}}(S^3)$ are therefore $\mathbb{Q}$-vector spaces as well. The canonical map $\widetilde{\mathcal{P}}(S^3) \to \widetilde{\mathcal{P}}(S^3) \otimes \mathbb{Q}$ is injective, so $\bar{z} = 0$. □

This proof is only a composition of functions, preservation of zero, and injectivity. Lean proves it parametrically from the fields of the external package; it does not use the names of the geometric theories to bypass any arrow.

5.4. **Removing suspension.** Because $\bar{z} = 0$ in the cokernel of suspension, there is $Q \in \widehat{\mathcal{P}}(S^2)$ such that

$$z = \Sigma Q. \tag{5.5}$$

Input (E8) includes the suspension-volume identity. Its normalization can be seen on a polygon directly: suspension from the two poles has warped-product

volume element $\cos^2(t)\,dt\,dA$, where $-\pi/2 \le t \le \pi/2$. Hence

$$\mathrm{Vol}(\Sigma Q) = \left(\int_{-\pi/2}^{\pi/2} \cos^2 t\,dt\right) \mathrm{Area}(Q) = \frac{\pi}{2}\,\mathrm{Area}(Q). \tag{5.6}$$

Both volume and area are additive under cutting, so this formula extends from polygons to every class in $\widehat{\mathcal{P}}(S^2)$. The Lean proof does not formalize this integration; it consumes the resulting additive identity as the named field `suspension_volume` of Input (E8).

By Input (E1), Regge mates have equal volume. Applying volume to Equation (5.5) and using (5.6) gives $\mathrm{Area}(Q) = 0$. Input (E8) says that area is injective on $\widehat{\mathcal{P}}(S^2)$; hence $Q = 0$, so $z = 0$ in $\widehat{\mathcal{P}}(S^3)$. Its cancellation clause turns this equality into a finite equidecomposition. This proves Theorem 1.1.

5.5. **Logical inventory.** The last three sections now contain no hidden motivic inference. The internal implications are:

$$H^t H = 4I \Longrightarrow \text{four moving tensors agree},$$
$$\overline{\Delta}_U(\delta_U^\vee) = 0 \;\&\; \ker \overline{\Delta}_U \cong 0 \Longrightarrow \delta_U^\vee = 0,$$
$$\delta_U^\vee = 0 \Longrightarrow c_G(\delta_{s_T}^\vee) = 0 \Longrightarrow \bar{z} = 0 \Longrightarrow z = 0.$$

Every implication not reducible to this finite algebra or diagram chase is an explicitly numbered external input in Subsection 1.2.

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